A Dual-Branch Detail Extraction Network for Hyperspectral Pansharpening

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***Abstract*— Hyperspectral (HS) pansharpening aims at creating a high-resolution hyperspectral (HR-HS) image by integrating a high spatial resolution panchromatic (HR-PAN) image with a low-resolution hyperspectral (LR-HS) image. It is an important preprocessing procedure in many remote sensing tasks. Most of the existing pansharpening methods train a specific convolutional neural network (CNN) model for each type of dataset with the same number of spectral bands. The main contribution of this study is to propose a new dual-branch detail extraction pansharpening network (called DBDENet) that can sharpen HS images with any number of spectral bands using a single pre-trained model by fine-tuning the parameters of a small module in the network. Specifically, DBDENet extracts spatial details from LR-HS and HR-PAN images by two bidirectional branches of the dual-branch detail extraction network level by level. For each level, the spatial details captured from the HR-PAN and those of the LR-HS images are fused by a spatial cross attention fusion module (SCAFM). The spatial details fused by the last SCAFM module are injected into the upsampled HS image to obtain an HR-HS image. Experimental results prove to show the proposed DBDENet is superior to other widely accepted state-of-the-art methods in terms of objective indicators and visual appearance.**

***Index Terms*— Detail extraction, dual-branch, fusion, hyperspectral (HS) pansharpening.**

I. Introduction

HYPERSPECTRAL (HS) imaging can obtain dozens or even hundreds of dense sampled spectral bands about the same scene from visible to near-infrared light, which makes it possible to distinguish various materials with similar spectral signatures on the Earth. However, due to the limitation of cost and some physical conditions, the spatial resolution of HS images is relatively low [1]. HS images with low spatial performance do not meet the challenges posed by application tasks. HS pansharpening combines the spectral content of HS images and the spatial details of panchromatic (PAN) images to obtain pansharpened HS images with high spatial resolution and abundant spectral information. It provides an effective solution to overcome the disadvantage that a single sensor cannot satisfy both spectral and spatial requirements [2], [3]. In recent years, due to the increasing demand for high-resolution images in commercial platforms such as Google Map and Microsoft Bing and some remote sensing tasks such as anomaly detection and target tracking [4]–[7], HS pansharpening has attracted more attention in remote sensing image processing [8]–[10].

In the past few decades, many popular pansharpening algorithms have been proposed [11], [12]. They are mainly divided into the following four categories: classical component substitution (CS) and multiresolution analysis (MRA) pansharpening methods based on simple models, and recently proposed matrix factorization and Bayesian pansharpening methods based on complex models [13]–[17]. The CS and MRA pansharpening methods are simple and fast to implement, so they are popular in large real-time application systems. They can be represented by a unified model, that is, the whole pansharpening process can be divided into two separate processes of details extraction and details injection. The main difference between these two categories of pansharpening algorithms is in detail extraction. The CS pansharpening methods use a PAN image to replace the intensity component of the HS image, while the MRA methods directly extract the high-frequency component of PAN image to achieve pansharpening. The widely used CS pansharpening methods are Gram–Schmidt (GS) [18], adaptive GS (GSA) [19], intensity hue saturation (IHS) transform [20], principal component analysis (PCA) [21], [22] and the guided filter PCA (GFPCA) [23]. The spectral quality of pansharpened images provided by CS methods is not ideal due to the mismatch of the spectral range. Spectral range mismatch is an inherent problem with HS pansharpening, and this problem affects the performance of HS pansharpening methods. The CS pansharpening methods which use the PAN image to replace the intensity component of the HS image are greatly affected by the problem of spectral

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range mismatch and cause significant spectral distortion. At present, research on CS pansharpening methods mainly focus on the estimation of context-adaptive injection coefficients. Because of their excellent spectral performance, MRA pansharpening algorithms are still being investigated, concentrating on the accurate estimation of missing spatial details of HS images. Generally, missing details are estimated via an MRA, such as non-subsampled contourlets transform [21], curvelets transform [24], wavelet transform [25], [26], and Laplacian pyramid [27]. Compared with the CS methods, the MRA can usually achieve relatively higher spectral fidelity at the sacrifice of spatial quality.

The Bayesian and matrix factorization pansharpening methods recently proposed to rely on a regularized solution of an ill-posed problem. Bayesian naive Gaussian prior (Bayesian Naive) [28], HySure [29], and Bayesian sparsity promoted Gaussian prior (Bayesian Sparse) [30] are classical representations of Bayesian pansharpening methods. Although the Bayesian and matrix factorization pansharpening methods perform well in terms of spectral fidelity and spatial resolution compared to traditional pansharpening methods, they are time-consuming. Besides, it is difficult to find a suitable statistical model to jointly characterize the HS image, PAN image, and the pansharpened image, which greatly limits its application in pansharpening. To reduce the computational complexity, Wei *et al.* [30] presented a Sylvester equation-based pansharpening algorithm to reduce the time cost without significantly degrading the pansharpening performance. The coupled nonnegative matrix factorization (CNMF) which is originally proposed to achieve the fusion of HS and multispectral (MS) images is a popular unmixing-based pansharpening method [31]. However, how to make a tradeoff between time complexity and pansharpening performance remains a challenge for the Bayesian and matrix factorization pansharpening methods.

In recent years, deep neural networks, especially convolutional neural networks (CNNs), have been deeply explored in remote sensing image processing. Dong *et al.* [32] first developed a super-resolution CNN (SRCNN) model, which is a milestone for CNN in the field of image super-resolution. So far, various CNN architectures have been designed for image super-resolution. Pansharpening can be regarded as a special case of super-resolution, which uses the PAN and low-resolution hyperspectral (LR-HS) images to reconstruct high-resolution hyperspectral (HR-HS) images. As the advantages of CNN in image super-resolution, a number of CNN-based pansharpening methods began to emerge. Masi *et al.* [33] designed a three-layer deep network (PNN) to solve the pansharpening problem by using the idea from a super-resolution network. Inspired by PNN, Scarpa *et al.* [34] explored a variety of alternative architectures and learning modalities, and proposed a target-adaptive tuning strategy to solve the problem of insufficient training data. Wei *et al.* [35] introduced residual learning into CNN and developed a deep residual pansharpening neural network (DRPNN), which makes full use of the nonlinear representation of deep learning network to improve pansharpening performance. Yuan *et al.* [36] introduced multiscale extraction and residual learning into the CNN network, and proposed a multiscale and multidepth CNN-based pansharpening method. Ye *et al.* [37] proposed a gradient-based deep network prior for pansharpening, which integrated the gradient-based deep network prior into model-based optimization. He *et al.* [38] proposed a detailed injection-based CNN framework (DiCNN) with a clear physical explanation. It is worth mentioning that given the pre-trained DiCNN2, it only needs to fine-tune the last convolution layer of DiCNN2 to achieve the pansharpening of a new testing dataset with different number of bands, which is of great value for subsequent studies. The above introduces the successful application of CNN in MS pansharpening. In recent years, CNN-based HS pansharpening methods have also appeared. He *et al.* [39] proposed a spectrally predictive CNN-based HS pansharpening framework (HyperPNN), which introduced spectrally predictive structure to mitigate spectral distortion and spatial blurring. Li *et al.* [40] proposed a detail-based deep Laplacian HS pansharpening method (DDLPS), focusing on both spatial and spectral preservations. Dong *et al.* [41] proposed a Laplacian pyramid dense network (LPPNet) for HS pansharpening, which applied the Laplace pyramid decomposition technique to CNN to simplify the pansharpening problem into several pyramid-level learning problems, and made the network more efficient.

In this study, we present a promising dual-branch CNN pansharpening algorithm [dual-branch detail extraction pansharpening network (DBDENet)], which intends to estimate spatial details progressively. The major contributions of the proposed method can be summarized as follows.

1) Most of the traditional CNN-based methods were originally designed for super-resolution and they work like a black-box. The proposed model includes two continuous processes: detail extraction and image reconstruction, with a clear physical interpretation for HS pansharpening.
2) DBDENet extracts the details from LR-HS and HR-PAN images by two bidirectional branches and fuses the extracted details by a spatial cross attention fusion module (SCAFM), so as to make full use of the hierarchical features from LR-HS and HR-PAN images.
3) We make the DBDENet more general, that is, a unified network pre-trained on a certain kind of images can be used for pansharpening HS images with different numbers of spectral bands by finetuning the network parameters of the last layer in the second SCAFM.

The remainder of this study is organized as follows. Section II describes the proposed DBDENet pansharpening approach. The experimental results and analysis are given in Section III. Section IV concludes this article.

II. Proposed Method

The main concern of this article is to adaptively estimate the missing details of HS images. Specifically, the proposed method contains two continuous processes which are detail extraction and image reconstruction. The detail extraction subnetwork is achieved by a dual-branch fusion network, in which the details of HS and that of PAN images are

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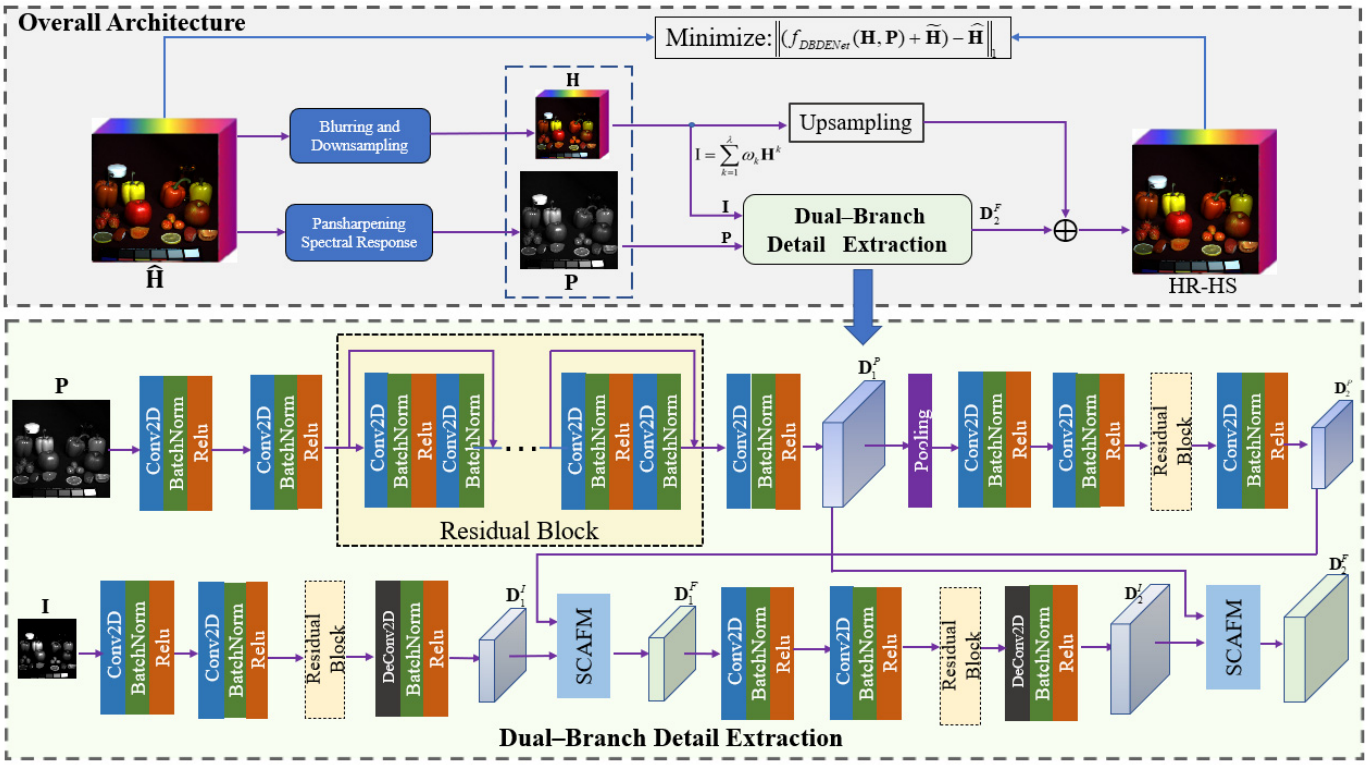


Fig. 1. HS pansharpening based on dual–branch detail extraction network.

extracted in two separate branches. The spatial component of the HS image is upsampled while the PAN image is downsampled correspondingly which makes the network more efficient. The extracted details of each level are fused by the SCAFM module, and the fused detail maps of the top level are then injected into the upsampled HS image to obtain the pansharpened HS image. The architecture of the proposed DBDENet is provided in Fig. 1. Let $\mathbf{H} \in \mathrm{R}^{m\times n\times\lambda}$ denote the acquired LR-HS, where λ denotes the total number of spectral bands in the HS image, and $m \times n$ stands for the number of pixels. $\widetilde{\mathbf{H}} \in \mathrm{R}^{M\times N\times\lambda}$ denotes the interpolated HS image of PAN size, $\widehat{\mathbf{H}} \in \mathrm{R}^{M\times N\times\lambda}$ represents the fused HR-HS image, where $M \times N = d^2(m \times n)$ is the pixel number in HR-HS image, and d is the up-sampling factor. The PAN image is denoted by $\mathbf{P} \in \mathrm{R}^{M\times N\times 1}$. $\widehat{\mathbf{H}}$ and $\mathbf{H}$ share the same number of spectral bands, $\widehat{\mathbf{H}}$ and $\mathbf{P}$ share the same number of pixels. The proposed DBDENet can be utilized to process the PAN images whose spatial resolution is 2^i times that of HS images, where i is the number of levels. This article takes four times as an example to explain in detail as follows.

A. Dual–Branch Fusion-Based Multilevel Residual Detail Extraction Subnetwork

The proposed method estimates the missing details by a dual-branch fusion-based multilevel residual detail extraction subnetwork, in which, the spatial component of HS image and PAN image are processed in two separate residual branches level by level, and the extracted details are fused by the SCAFM. In particular, the dual-branch fusion-based multilevel residual detail extraction subnetwork contains two branches: the PAN detail extraction branch and the detail fusion branch.

1) PAN Detail Extraction Branch: Residual network (ResNet) can extract more representative features with deeper CNN and solve the gradient vanishing problem. Some ResNet-based pansharpening methods [34]–[36] are proposed recently, which demonstrate the superiority of ResNet in solving pansharpening problem. Since the available LR-HS images are similar to the desired HR-HS images, the ResNet can generate residual images with most of the values being zero or close to zero. Based on the ResNet, high nonlinearities can be exploited and high-level features can be provided. Therefore, it is an effective way to adopt the ResNet for pansharpening.

The main task of PAN detail extraction branch is to extract the spatial component of PAN image level by level. The resolution of the desired HR-HS image with high spatial quality is generally four times that of the available LR-HS image. Therefore, in this example, the PAN detail extraction branch consists of two levels. Each level of the PAN detail extraction branch starts with two convolutional layers, followed by a Resblock module, and finally a convolutional layer.

For the first level of the detail extraction branch, the PAN image is fed into the convolution layers to generate a 64-channel feature map. The residual features are then extracted by the Resblocks. In order to convert the multichannel feature map into a fixed-channel detail image, the output of the ResNet is fed into a convolutional layer. The output of this convolutional layer is the input of the second level, and it is also used to fuse the feature map obtained by the second level of the HS detail extraction branch. The first level of the

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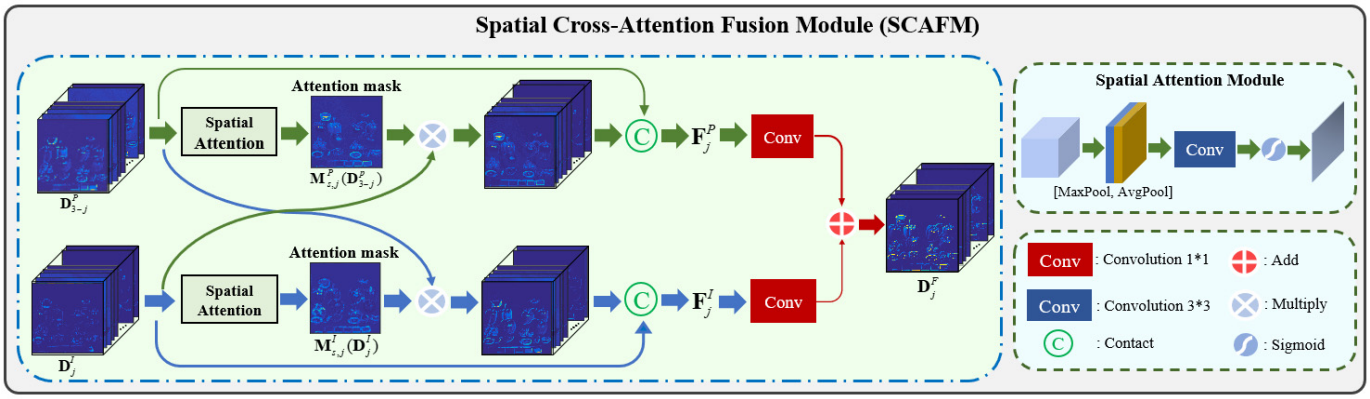


Fig. 2. Structures of the SCAFM module.

PAN detail extraction branch can be given by the following formulation:

$$\mathbf{D}_1^P = f_{\text{conv},13}\left\{\underbrace{f_{res,11},\ldots,f_{\text{res},1t}}_{\times t}\left[f_{\text{conv},12}\left(f_{\text{conv},11}(\mathbf{P})\right)\right]\right\} \quad (1)$$

where subscript 1 indicates the level index, $\mathbf{D}_1^P$ denotes the output of the first level, $f_{\text{conv},11}$, $f_{\text{conv},12}$, and $f_{\text{conv},13}$ denote the initial two convolution layers and the last convolution layer of the first level in the PAN detail extraction branch, respectively, $f_{\text{res},1t}$ denotes the tth Resblock of the first level in the PAN detail extraction branch, $\times t$ denotes that the features are processed by t Resblocks, and t is the number of Resblocks in the first level.

There are two differences between the first and second levels. First, the input of the second level needs to go through an average-pooling layer to downsample the feature map to the size of the LR-HS image. Second, the output of the second level is fused with the output of the first level of the HS detail extraction branch to reconstruct the spatial component. The second level of the PAN detail extraction branch can be given by the following formulation:

$$\mathbf{D}_2^P = f_{\text{conv},23} \times \left\{\underbrace{f_{res,21},\ldots,f_{\text{res},2t}}_{\times t}\left[f_{\text{conv},22}\left(f_{\text{conv},21}\left(\text{pooling}\left(\mathbf{D}_1^P\right)\right)\right)\right]\right\} \quad (2)$$

where, similarly, subscript 2 indicates the level index, $\mathbf{D}_2^P$ denotes the output of the second level, pooling denotes the pooling operation, $f_{\text{conv},21}$, $f_{\text{conv},22}$, and $f_{\text{conv},23}$ denote the initial two convolution layers and the last convolution layer of the second level in the PAN detail extraction branch, respectively, and $f_{\text{res},2t}$ denotes the tth Resblock of the second level.

2) Detail Fusion Branch: The task of the detail fusion branch includes extracting the spatial feature maps of the HS image and fusing them with the feature maps obtained by the PAN detail extraction branch. Same as the PAN detail extraction branch, the detail fusion branch also contains two levels. Each level of the detail fusion branch contains two convolutional layers, a Resblock module and a deconvolutional layer.

At the first level of the detail fusion branch, the spatial component of the HS image $\mathbf{I}$ is fed into a convolutional layer that contains 64 filters with the size of 3×3. The spatial component $\mathbf{I}$ is extracted by an adaptive weighted algorithm [19], which can be described by

$$\mathbf{I} = \sum_{k=1}^{\lambda} \omega_k \mathbf{H}^k$$
$$\{\omega_k\}_{k=1,\ldots,\lambda} = \arg\min_{\{\omega_k\}_{k=1,\ldots,\lambda}} \|\sum_{k=1}^{\lambda} \omega_k \mathbf{H}^k - \downarrow \mathbf{P}\|^2 \quad (3)$$

where ω_k is the combination coefficient, and $\downarrow$ is the downsampling operation. The second convolutional layer contains 64 filters with the size of 3×3. The detail fusion branch keeps the high-level architecture design of the PAN detail extraction branch, and adopts a Resblock as illustrated in Fig. 1. When the feature map provided by the Resblock is fed into the deconvolutional layer, a feature map with 128 channels is created. The output of the deconvolutional layer is an interpolated feature map that is doubled in size. The produced output feature maps at the first level can be expressed as

$$\mathbf{D}_1^I = g_{\text{deconv},13}\left\{g_{\text{Resblock1},t}\left[g_{\text{conv},12}\left(g_{\text{conv},11}(\mathbf{I})\right)\right]\right\} \quad (4)$$

where subscript 1 indicates the level index, $\mathbf{D}_1^I$ is the output feature maps of the first level, $g_{\text{conv},11}$ and $g_{\text{conv},12}$ are the initial two convolution layers at the first level of the detail fusion branch, $g_{\text{Resblock1},t}$ denotes the Resblock module with t Resblocks, and $g_{\text{deconv},13}$ denotes the deconvolutional layer of the first level.

The details extracted from the HR-PAN and LR-HS images are fused at each level. As shown in Fig. 1, in order to further exploit the significant spatial information from two branches, the SCAFM module which is a flexible component is embedded between each level of the two branches. The SCAFM module captures the region of interest, highlights the most advantageous spatial information, and transfers the significant spatial information across the two branches to yield more effective spatial information transfer in network. The SCAFM module is illustrated in Fig. 2. As can be seen, the feature map $\mathbf{D}_j^I$ provided the jth level of detail fusion branch and the feature map $\mathbf{D}_{3-j}^P$ produced by the $(3-j)$th level of

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detail extraction branch are served as the inputs of the SCAFM module. Since $\mathbf{D}_j^I$ and $\mathbf{D}_{3-j}^P$ can provide more faithful spatial guidance, we first calculate the attention masks $\mathbf{M}_{s,j}^I(\mathbf{D}_j^I)$ and $\mathbf{M}_{s,j}^P(\mathbf{D}_{3-j}^P)$ from $\mathbf{D}_j^I$ and $\mathbf{D}_{3-j}^P$, respectively. To transfer the advantageous information across two branches, the obtained attention masks are then multiplied with the input feature map of another branch. The obtained cross-multiplications are finally concatenated with the original input features for each branch, to enrich the spatial information. This procedure can be formulated as

$$\begin{aligned}\mathbf{F}_j^P &= \text{concat}\left[\mathbf{D}_{3-j}^P, \mathbf{D}_j^I \otimes \mathbf{M}_{s,j}^P\left(\mathbf{D}_{3-j}^P\right)\right], \quad j = 1, 2\\ \mathbf{F}_j^I &= \text{concat}\left[\mathbf{D}_j^I, \mathbf{D}_{3-j}^P \otimes \mathbf{M}_{s,j}^I\left(\mathbf{D}_j^I\right)\right], \quad j = 1, 2\end{aligned} \tag{5}$$

where $\mathbf{F}_j^P$ and $\mathbf{F}_j^I$ are the concatenated features of the two branches at the SCAFM module of the jth level, concat[ ] denotes the concatenation operation, $\otimes$ represents element-wise multiplication, and $\mathbf{M}_{s,j}^P(\cdot)$ and $\mathbf{M}_{s,j}^I(\cdot)$ denote spatial attention masks of the two branches at jth level. For the first level, the concatenated features $\mathbf{F}_1^I$ and $\mathbf{F}_1^P$ are fed into a convolutional layer that contains 128 filters with the size of 1×1. The output maps of two branches are added to obtain the fused detail map of the first level $\mathbf{D}_1^F$, which is given as

$$\mathbf{D}_1^F = f_{\text{conv},14}\left(\mathbf{F}_1^P\right) + g_{\text{conv},14}\left(\mathbf{F}_1^I\right) \tag{6}$$

where $f_{\text{conv},14}$ and $g_{\text{conv},14}$ stand for the convolutional operations of the two branches at the first level.

Except for the inputs and outputs, the structure of the second level of the detail fusion branch is highly consistent with that of the first level. The inputs of the second level $\mathbf{D}_2^I$ can be expressed as

$$\mathbf{D}_2^I = g_{\text{deconv},23}\left\{g_{\text{Resblock2},t}\left[g_{\text{conv},22}\left(g_{\text{conv},21}\left(\mathbf{D}_1^F\right)\right)\right]\right\} \tag{7}$$

where $g_{\text{deconv},23}$ is the deconvolutional layer of the second level, $g_{\text{conv},21}$ and $g_{\text{conv},22}$ are the two convolution functions, $g_{\text{Resblock2},t}$ is the Resblock module at the second level of the detail fusion branch. For the second level of the details fusion branch, the concatenated features $\mathbf{F}_2^I$ and $\mathbf{F}_2^P$ are also fed into an SCAFM module to generate the fused detail map $\mathbf{D}_2^F$, which can be formulated as

$$\mathbf{D}_2^F = f_{\text{conv},24}\left(\mathbf{F}_2^P\right) + g_{\text{conv},24}\left(\mathbf{F}_2^I\right) \tag{8}$$

where $f_{\text{conv},24}$ and $g_{\text{conv},24}$ denote the convolutional layer that contains λ filters with the size of 1×1, in which λ is equal to the number of bands in the HS image. Extensive experimental results demonstrate that it is possible to use the pre-trained DBDENet model to sharpen the HS images with arbitrary number of spectral bands only by fine-tuning the network parameters in the last convolutional layer of the second SCAFM module.

B. Pansharpened HS Image Reconstruction

Following the general idea of MRA, the reconstructed residual map of the second level is injected into the upsampled HS image $\widetilde{\mathbf{H}}$ to generate the fused HS image $\widehat{\mathbf{H}}$. This process can be formulated as

$$\widehat{\mathbf{H}} = \widetilde{\mathbf{H}} + \mathbf{D}_2^F \tag{9}$$

where $\mathbf{D}_2^F$ is the reconstructed residual map by the dual–branch fusion-based multilevel residual detail extraction subnetwork.

C. Loss Function

Let $\mathbf{H}$, $\mathbf{P}$ and $\mathbf{F}$ represent the LR-HS image, the HR-PAN image and the label HS image, respectively. The training set is expressed as $\{\mathbf{H}^{(m)}, \mathbf{P}^{(m)}, \mathbf{F}^{(m)}\}_{m=1}^M$, where M denotes the sample size. The objective of the training network is to predict the residual image $\mathbf{D}_2^F$, which is expressed as

$$\mathbf{D}_2^{F(m)} = f_{\text{DBDENet}}\left(\mathbf{H}^{(m)}, \mathbf{P}^{(m)}\right) \tag{10}$$

where $m = 1, 2, \ldots, M$, $\mathbf{D}_2^{F(m)}$ represents the mth sample of the predicted residual image, and f_{DBDENet} represents the proposed DBDENet network. The L_1 loss function is utilized to optimize the proposed DBDENet network. The loss function is described as

$$L = \frac{1}{M}\sum_{m=1}^{M} \left\| f_{\text{DBDENet}}\left(\mathbf{H}^{(m)}, \mathbf{P}^{(m)}\right) - \left(\widehat{\mathbf{H}}^{(m)} - \widetilde{\mathbf{H}}^{(m)}\right)\right\|_1. \tag{11}$$

III. Experimental Results and Analysis

A. Datasets and Experimental Setup

In order to evaluate the effectiveness of the proposed DBDENet method, experiments are conducted on three datasets which are the Pavia Center dataset, Columbia Computer Vision Laboratory (CAVE) dataset, and the Botswana dataset. The first dataset is the Pavia Center dataset[1] which was collected via the reflective optics system imaging spectrometer (ROSIS). The ROSIS provides 115 bands of data in a spectral range of 0.4–0.9 μm, and 102 bands are used for the experiments after removing the water vapor absorption and noise bands. The Pavia Center is with 1096×1096 pixels in the spatial aspect. However, pixels not containing any information are discarded, leaving 1096×715 spatial pixels for experimentation. We intercepted the part of size $960 \times 600 \times 102$ for the experiments. The obtained part are divided into 558 patches of size $120 \times 120 \times 102$ with stride 28. These HS patches are used as the reference HS images. The test HS images with $30 \times 30 \times 102$ pixels and the test PAN images of size 120×120 are generated according to Wald's protocol [42]. For the Pavia Center data, the first 391 HS and PAN pairs (about 70%) are used for training, and the last 167 pairs are for testing.

The second dataset is the CAVE dataset[2] consisting of 32 indoor HS images. These HS images were acquired by the generalized assorted pixel camera. The CAVE dataset has 31 bands in the spectral range of 0.4–0.7 μm. All of these images have a size of 512×512. The available HS images are served as reference images. The low-resolution HS images and the high-resolution PAN images are generated according to Wald's protocol [42]. Specifically, the PAN images are generated by averaging the visible bands of the reference images. The simulated HS images are obtained by applying a 9×9 Gaussian

[1] https://rslab.ut.ac.ir/data

[2] https://www1.cs.columbia.edu/CAVE/databases/multispectral/

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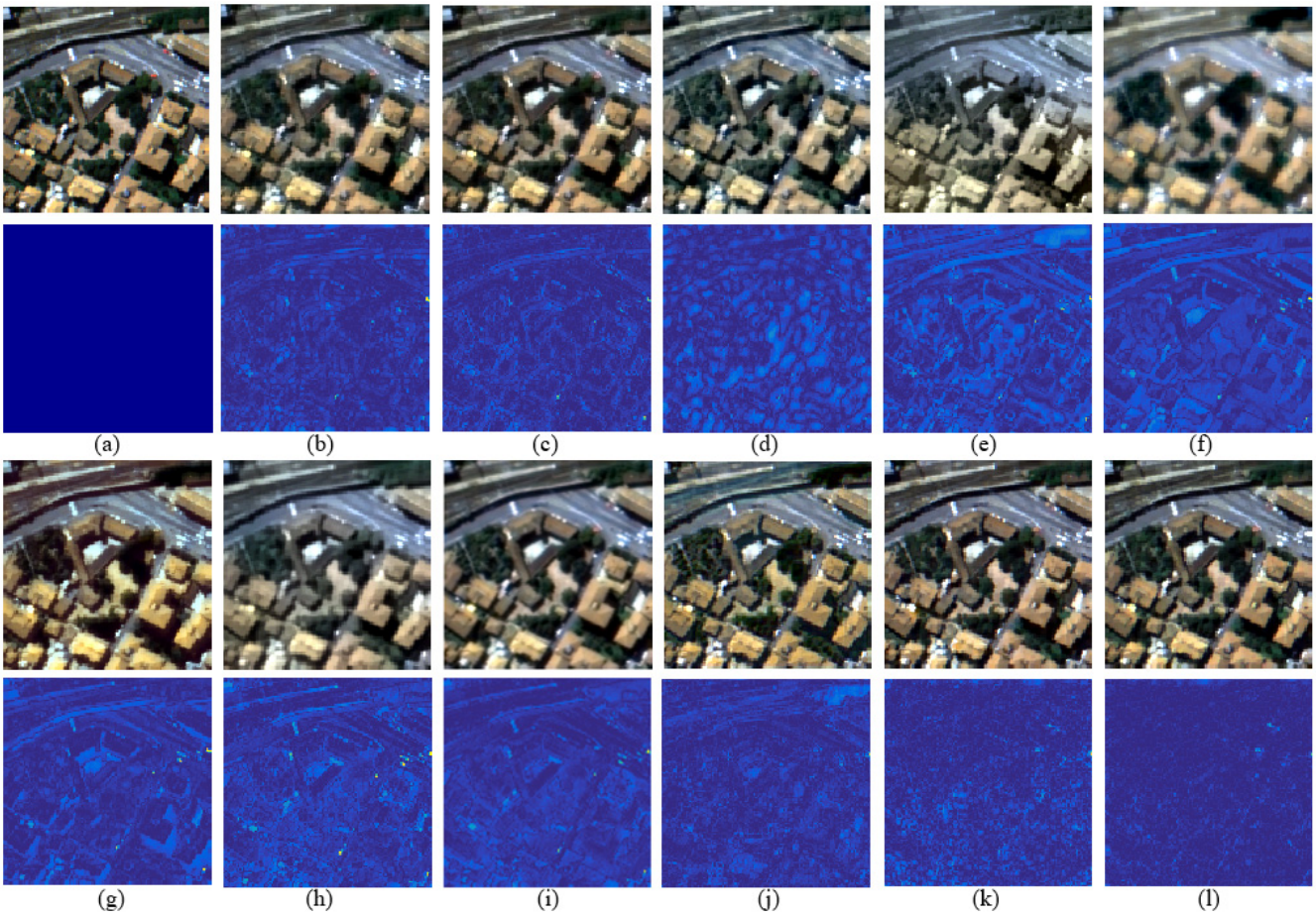


Fig. 3. Visual results of Pavia Center dataset. (a) Reference HS image. (b) SFIM. (c) MGH. (d) GSA. (e) PCA. (f) GFPCA. (g) CNMF. (h) Bayes. (i) DDLPS. (j) HyperPNN. (k) DiCNN2. (l) DBDENet. The odd rows show the pansharpening images of each method, and the even rows show the error images of the corresponding method.

filter and then downsampling with a scale factor of 4. Thus, the dimension of the simulated HS image is 128×128 pixels, whereas the size of the PAN image is 512×512. 22 HS images (about 70%) are selected for training, and the remaining ten HS images are used for testing. For the CAVE data training, each reference HS image, low resolution HS image and PAN image are divided into 16 patches of size $128 \times 128 \times 31$, $32 \times 32 \times 31$ and 128×128 without overlapping, respectively.

The Botswana dataset[3] is the last data which was acquired by the Hyperion instrument on the Earth observing one (EO-1) spacecraft. Hyperion instrument is capable of providing HS images of 242 bands within the spectrum range of 0.4–2.5 μm. The uncalibrated and noisy bands are removed, and 145 bands are used for experimentation. The Botswana data were originally with 1476×256 pixels, and the part of 1200×240 pixels was captured for the experiments. We have divided the captured part into 544 cubic patches with $120 \times 120 \times 145$ pixels with stride 16. For the Botswana dataset, 544 HS cubic-patches are served as the reference HS images, and the test HS and PAN images are $30 \times 30 \times 145$ pixels with a spatial resolution of 30 m and 120×120 pixels with a spatial resolution of 10 m, respectively. The first 380 HS and PAN pairs (about 70%) are selected for training, and the remaining 164 pairs are used for testing.

The model of the proposed method is trained on the Pavia Center dataset, and the trained model is applied on the CAVE and Botswana datasets by fine-tuning the parameters of the last convolution layer of the second SCAFM module. Training and testing of the proposed method are implemented based on the PyTorch framework using the NVIDIA GeForce GTX 2080Ti GPU. The same hyperparameters are set for the three datasets. The batch size is set to 8, and the training process terminates at 500 epochs. The Adam optimizer is adopted to minimize the loss function, and the initial learning rate is set to 5×10^{-4}. The learning rate descent factor is set to 0.1 every 200 epochs. Code is available at https://github.com/Jiahuiqu/DBDENet.

B. Compared Methods and Quality Assessment

For the purpose of comparison, the proposed DBDENet method is compared with ten widely accepted HS pansharpening methods, namely smoothing filter-based intensity modulation (SFIM) [43], modulation transfer function-generalized Laplacian pyramid with high pass modulation (MGH) [11], GSA [19], PCA [20], GFPCA [23], CNMF [31], Bayes [30], DDLPS [40], HyperPNN [39], and DiCNN2 [38]. Among these compared methods, DDLPS, HyperPNN, and DiCNN2 are all recently proposed the CNN-based state-of-the-art pan-

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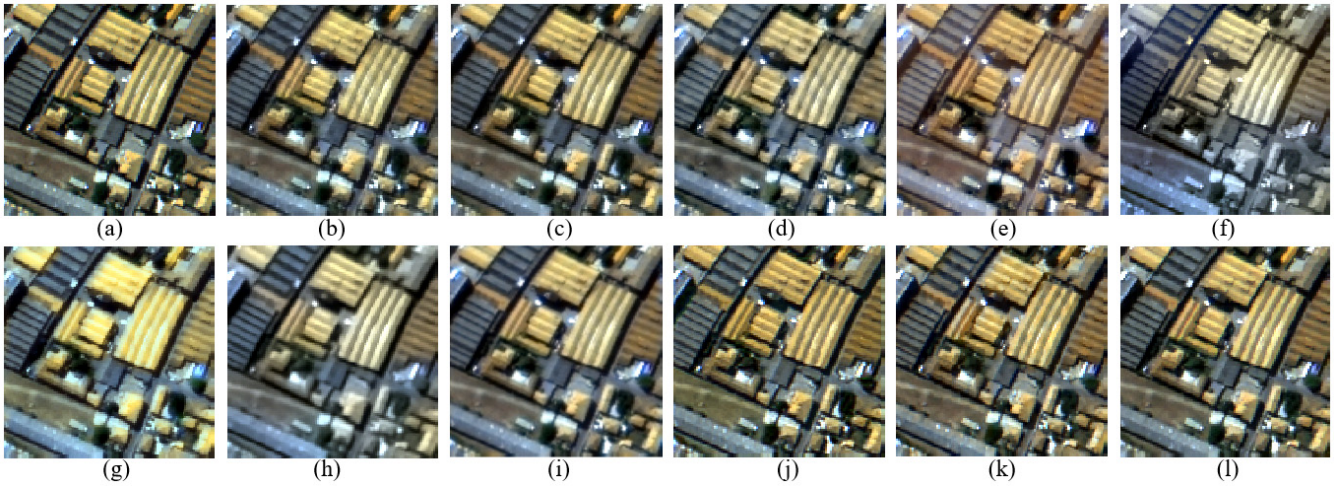


Fig. 4. Pansharpening results of another HS image from the Pavia Center dataset. (a) Reference HS image. (b) SFIM. (c) MGH. (d) GSA. (e) PCA. (f) GFPCA. (g) CNMF. (h) Bayes. (i) DDLPS. (j) HyperPNN. (k) DiCNN2. (l) DBDENet.

sharpening algorithm. For the DiCNN2 and the proposed DBDENet method, once the network is trained, the parameters in the network are fixed, except for the parameters of the last layer. Specifically, for a new test dataset, only the last convolution layer of DiCNN2 and the last convolution layer of the second SCAFM module in the proposed DBDENet method need to be fine-tuned. The models of the DiCNN2 and the proposed algorithm are all trained on the Pavia Center dataset, and the trained models are performed on the CAVE and Botswana datasets just with slight finetuning. For a fair comparison, we applied the same convolutional layer settings to DiCNN2 and DBDENet. We tune the parameters so that all the models can obtain their best performance. SFIM and MGH are two typical MRA-based methods and usually have good spectral preservation performance. GSA and PCA are known for their fast and simple implementation. GFPCA is a hybrid method, and won the "Best Paper Challenge" award at the 2014 IEEE Data Fusion Contest. CNMF is based on matrix factorization, and Bayes is a representative algorithm of the Bayesian class. These compared algorithms cover the latest effective work and various categories reviewed in the Introduction.

To evaluate the fusion performance, the widely used objective indexes and the visual appearance are considered. We first generate the degraded HS image and high-resolution PAN image according to Wald's protocol. Then, different pansharpening methods are performed on the degraded HS image and high-resolution PAN image to generate a fused image. Finally, the fused image is compared with the available HR-HS image to evaluate pansharpening performance using several widely accepted objective indexes, including cross correlation (CC) [44], spectral angle mapper (SAM), root mean squared error (RMSE), and Erreur Relative Globale Adimensionnelle de synthèse (ERGAS) [45]. CC is a spatial index calculated by averaging the CC value of each band and is used to assess the correlation degree between a pansharpened image and the reference image. SAM takes the average value of the difference image and reflects the spectral distortion caused by the pansharpening algorithm by measuring the distortion degree between the spectral vector of the pansharpening result and that of the reference image. Both RMSE and ERGAS are global metrics used to account for the spectral and spatial quality of fused images. The closer the value of CC is to 1, the closer the values of SAM, RMSE, and ERGAS are to 0, the better the spatial and spectral performance of the pansharpened image.

C. Experimental Results

For the Pavia Center dataset, the visual results obtained by different methods are shown in Fig. 3. With respect to visual appearance, all generated images suffer from more or less global distortion, which is reflected by luminance. The images generated by DDLPS and the proposed DBDENet method include the least distortion. Images generated by SFIM, MGH, GFPCA, and Bayes are blurry in object boundaries, which suggests that these pansharpening methods fail to restore the details of the original HS image properly. Though CNMF and HyperPNN show a great ability of detail preservation, they suffer from severe spectral distortion as indicated by the roof areas of buildings and the road. GSA demonstrates great fusion ability in both details and global appearance, but GSA cannot perform well in certain materials, e.g., a curved road in Pavia Center. The visual results of Bayes and DDLPS are not as good as the proposed method and DiCNN2, and they have a little spectral distortion. In order to further compare the performance of each method, the error images of each method for the Pavia Center dataset are shown in the even rows of Fig. 3. Yellow means large differences, and blue means small differences. The DBDENet method shows the smallest differences in most regions, which testifies the superior performance of the proposed method.

Table I reports the values of objective measures for the Pavia Center dataset. Compared with the competing methods, the proposed DBDENet yields higher values of CC and lower values of SAM, RMSE, and ERGAS, indicating that the proposed DBDENet method outperforms.

Fig. 4 shows the pansharpening results of another HS image from the Pavia Center dataset. Similar to the previous

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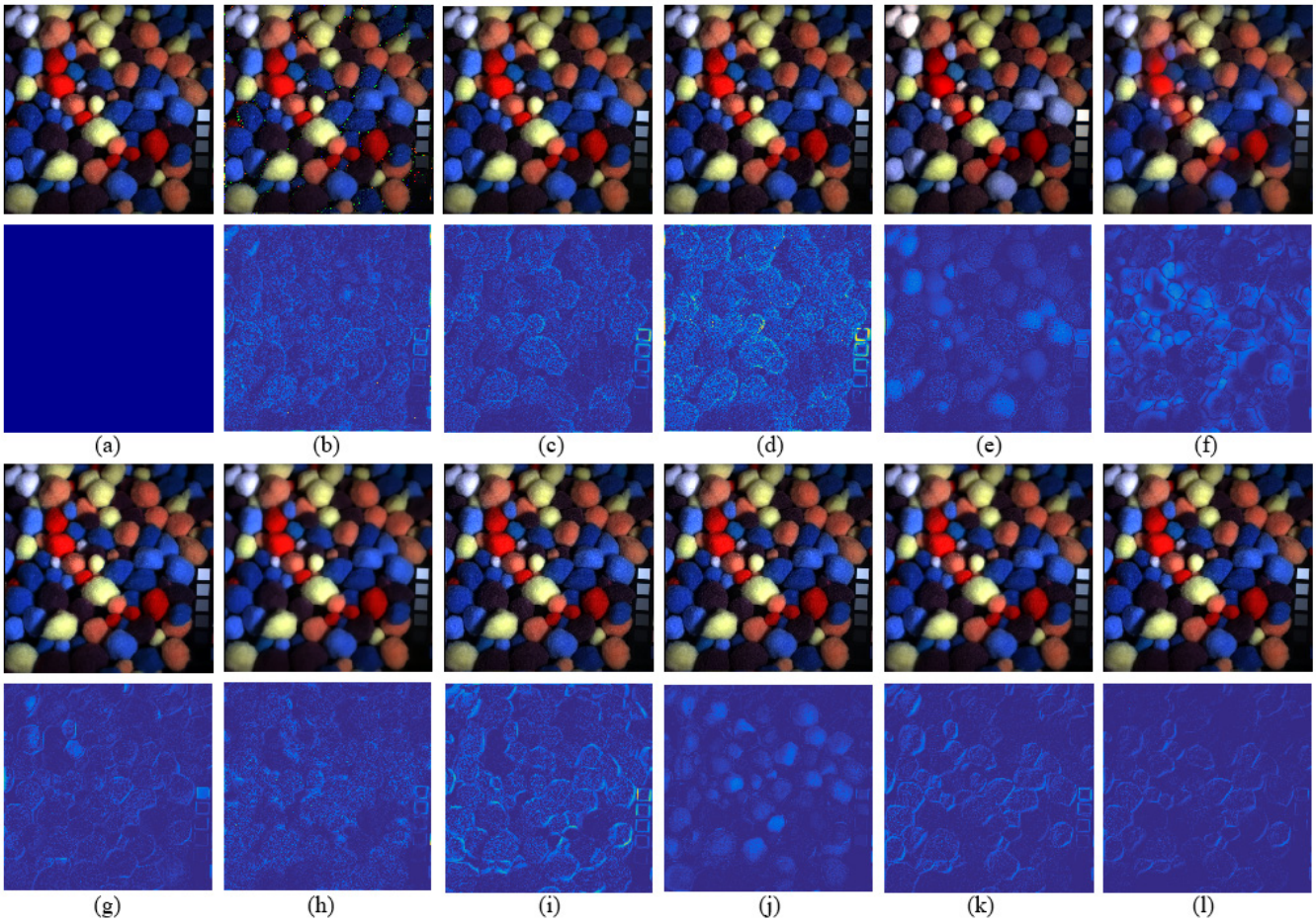


Fig. 5. Visual results of CAVE dataset. (a) Reference HS image. (b) SFIM. (c) MGH. (d) GSA. (e) PCA. (f) GFPCA. (g) CNMF. (h) Bayes. (i) DDLPS. (j) HyperPNN. (k) DiCNN2. (l) DBDENet. The odd rows show the pansharpening images of each method, and the even rows show the error images of the corresponding method.

TABLE I
AVERAGE QUANTITATIVE ASSESSMENT FOR THE PAVIA CENTER DATASET

Method	CC	SAM	RMSE	ERGAS
SFIM	0.8767	7.6691	0.0413	6.7313
MGH	0.9302	7.8699	0.0318	4.8931
GSA	0.9142	7.6115	0.0353	5.2960
PCA	0.8522	8.5776	0.0440	7.0963
GFPCA	0.8562	8.3991	0.0497	7.9692
CNMF	0.9034	7.6625	0.0375	6.0057
Bayes	0.9191	8.0932	0.0356	5.8430
DDLPS	0.9308	7.9683	0.0378	6.0152
HyperPNN	0.9627	6.2873	0.0228	4.4093
DiCNN2	0.9618	5.7515	0.0220	3.5072
DBDENet	**0.9704**	**5.1321**	**0.0193**	**3.0070**

experiment, most of the pansharpening methods are able to restore spatial details and spectral information. Regarding spectral fidelity, SFIM, MGH, DDLPS, and the proposed DBDENet show the best performance. Spectral distortion is introduced by the GSA, PCA, GFPCA, Bayes, HyperPNN, and DiCNN2 methods as indicated by roof color. Regarding spatial quality, the proposed DBDENet provides the highest quality result. The stripes of the roof are not recognizable in the SFIM, CNMF, and Bayes results indicating that these pansharpening methods fail to inject the detailed information properly. The visual comparison suggests that the proposed DBDENet provides the best performance in both spectral fidelity and spatial quality.

The results of different fusion methods on the CAVE dataset are shown in Fig. 5. The odd rows show the pansharpening results of each method, and the even rows show the error images of the corresponding method. Comparing the degraded HS image with the results of the compared methods, it can be seen that these pansharpening methods improve the spatial resolution of degraded HS image to some extent. The MGH and DBDENet pansharpening methods are able to provide a higher spatial resolution pansharpened image and can better preserve the spectral information of the source image than other methods. Comparing the reference high-resolution image with the fused result of SFIM, it can be observed that some regions of this fusion result are over-sharpened. Obviously, the SFIM method produces artifacts, and the oversharpening exists on the edge of the ball. Although the HyperPNN method has good fidelity of the spatial information, the HyperPNN method causes spectral distortion in some areas. Some regions of the fusion results of PCA and GFPCA are blurred due to insufficient detail injection. The spatial distortion caused by the CNMF and, to a minor extent, the GSA and the DDLPS approach is relatively obvious. The edge areas of

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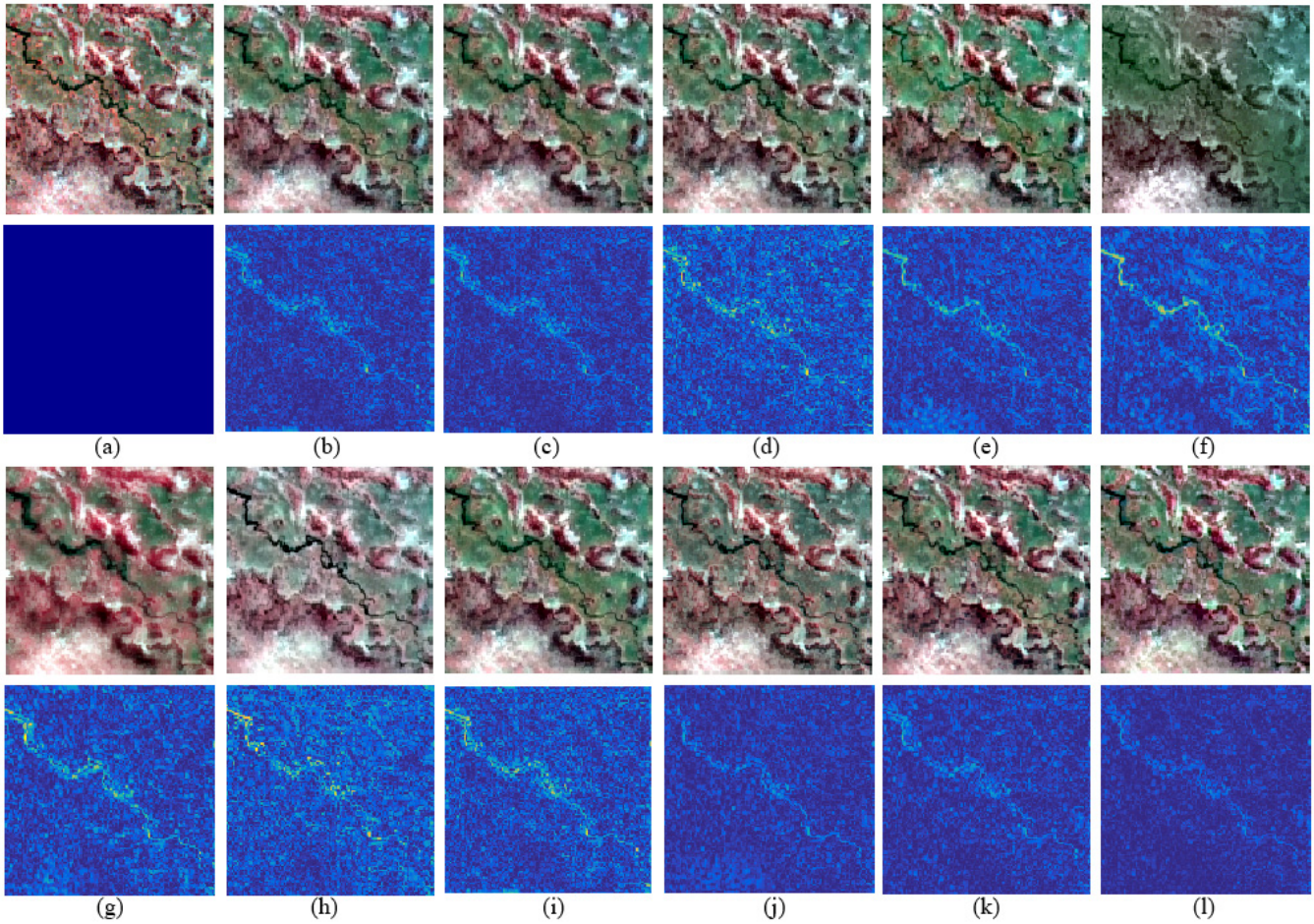


Fig. 6. Visual results of Botswana dataset. (a) Reference HS image. (b) SFIM. (c) MGH. (d) GSA. (e) PCA. (f) GFPCA. (g) CNMF. (h) Bayes. (i) DDLPS. (j) HyperPNN. (k) DiCNN2. (l) DBDENet. The odd rows show the pansharpening images of each method, and the even rows show the error images of the corresponding method.

the Bayes and DiCNN2 fusion result are consistent with that of the reference image, but spatial distortion is introduced in nonedged areas. The primary cause of these phenomena is that each band of HS image in these algorithms shares a single injection coefficient, which is disadvantageous for preserving the contrast of spatial details in PAN images. By comparing these pansharpening results in Fig. 5, we can observe that the details estimated by the proposed method turn out to be particularly beneficial to balance sharpening and blurring. The spatial improvement can be achievable with the help of employing the proposed dual-branch detail extraction network. This illustrates the proposed DBDENet pansharpening method has the ability to provide an HR-HS image that looks closest to the reference one. The superior extent of the fused result shows that the fusion performance is surely improved by the proposed method. Fig. 5 further shows the error images (absolute values) between each fused HS image and reference HS image of the CAVE dataset, and it can be seen that the proposed DBDENet method generates the HS image with the least error.

The objective evaluation of the fusion methods on the CAVE dataset shown in Fig. 5 is listed in Table II, where optimal values are in bold. Consistent with subjective analysis, the proposed DBDENet pansharpening method significantly improves the performance of the degraded LR-HS image and provides the optimal values of all the quantitative indexes, which indicates that the DBDENet method improves both spatial and spectral quality of the fused image. This further demonstrates the effectiveness of the proposed method in HS pansharpening.

TABLE II
AVERAGE QUANTITATIVE ASSESSMENT FOR THE CAVE DATASET

Method	CC	SAM	RMSE	ERGAS
SFIM	0.9781	6.1743	0.0307	7.3068
MGH	0.9865	6.9261	0.0221	5.5904
GSA	0.9919	6.2234	0.0180	4.3743
PCA	0.9862	7.4560	0.0259	4.9309
GFPCA	0.9874	4.4150	0.0238	5.3594
CNMF	0.9923	6.9245	0.0169	3.7384
Bayes	0.9924	5.5662	0.0168	3.9812
DDLPS	0.9925	4.3592	0.0189	3.8213
HyperPNN	0.9944	6.8523	0.0174	3.9695
DiCNN2	0.9914	6.9094	0.0194	4.5636
DBDENet	**0.9963**	**3.7448**	**0.0123**	**2.7254**

For the Botswana dataset, the false color images generated by the competitive methods are shown in the odd rows of Fig. 6. Obviously, the PCA, GFPCA, CNMF, and Bayes

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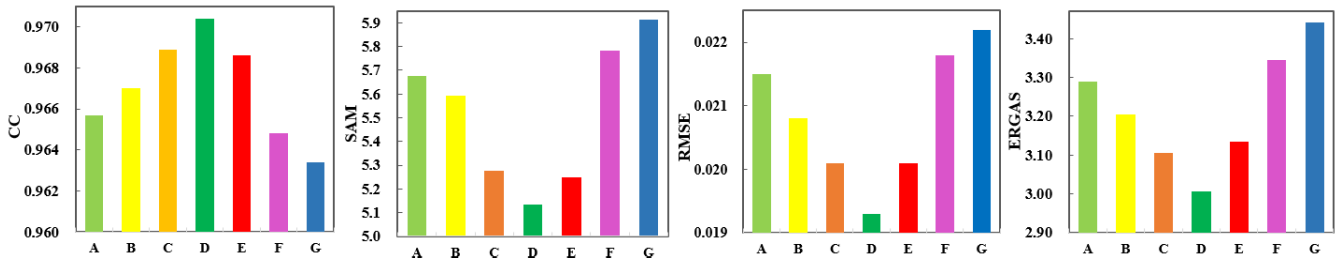


Fig. 7. Average quantitative assessment results for different models on the Pavia Center dataset. Dual-branch with Resblocks: (A) three Resblocks, (B) four Resblocks, (C) five Resblocks, (D) six Resblocks, (E) seven Resblocks, (F) only each level of PAN branch with six Resblocks, and (G) dual-branch without Resblocks.

TABLE III
AVERAGE QUANTITATIVE ASSESSMENT FOR THE BOTSWANA DATASET

Method	CC	SAM	RMSE	ERGAS
SFIM	0.9209	3.9174	0.0232	3.3340
MGH	0.9353	3.7267	0.0201	2.6656
GSA	0.9181	3.6905	0.0220	3.0694
PCA	0.9164	4.5715	0.0247	3.4976
GFPCA	0.9060	4.1958	0.0252	3.7686
CNMF	0.9188	3.4241	0.0204	2.9395
Bayes	0.9176	4.4829	0.0221	3.4571
DDLPS	0.9218	3.0154	0.0242	3.4016
HyperPNN	0.9383	1.8896	0.0115	2.3402
DiCNN2	0.9345	1.9536	0.0120	1.6562
DBDENet	**0.9453**	**1.8337**	**0.0107**	**1.5229**

TABLE IV
AVERAGE QUANTITATIVE ASSESSMENT OF DBDENET WITH FINE-TUNING AND DBDENET WITH TRAINING FROM SCRATCH

Data set	CAVE data set			
Method	CC	SAM	RMSE	ERGAS
DBDENet_from scratch	0.9962	3.6831	0.0127	2.7920
DBDENet_fine-tuning	0.9963	3.7448	0.0123	2.7254
Data set	Botswana data set			
Method	CC	SAM	RMSE	ERGAS
DBDENet_from scratch	0.9445	1.7072	0.0099	1.5065
DBDENet_fine-tuning	0.9453	1.8337	0.0107	1.5229

methods achieve blurry results in some areas, which illustrates that spatial details in the PAN image are not sufficiently injected. The visual effect produced by the Bayes method is much brighter than the ground truth image, while the visual result obtained by the GFPCA method is much darker than the ground truth image. The DDLPS method suffers from spectral distortion. Visually, the result provided by the proposed DBDENet is the closest to the ground truth image. The proposed method cannot only reconstruct the spectral information of the original HS image but also estimate the spatial details of the complex region of the texture accurately. The error images of each method shown in the even rows of Fig. 6 indicate that the proposed DBDENet method achieves the smallest difference among the competing methods, which demonstrates the excellent performance of our proposal.

In Table III, the quantitative indices obtained by different pansharpening methods on the Botswana dataset are listed. The DBDENet yields lower values of SAM, RMSE, and ERGAS, and larger values of CC. Compared with the competing methods, DBDENet offers a remarkable advantage in CC, SAM, RMSE, and ERGAS, which indicates that the proposed method shows better spatial and spectral performance than others. From the expectation to the HS images, the DBDENet definitely performs the best.

To study the impact of finetuning training strategy on performance, we divide our model structures into two types, i.e., DBDENet_from scratch and DBDENet_fine-tuning. We test these two models on the CAVE and Botswana datasets, and train their parameters to obtain their best performance. The results are shown in Table IV. It can be seen that the results of fine-tuning are comparable to the results of training from scratch. Especially, for the CAVE dataset, the performance of fine-tuning is slightly better than the performance of training from scratch, and the DBDENet_fine-tuning yields higher values of CC and lower values of RMSE and ERGAS, which indicates that the fine-tuning training strategy of the proposed DBDENet method is acceptable and effective.

D. Ablation Study

1) Residual Block Number: Intuitively, if we adopt more residual blocks to construct the dual-branch detail extraction network, the performance will be improved. We have conducted an experiment on five models with residual blocks 3, 4, 5, 6, and 7 on the Pavia Center dataset. The average quantitative assessment results are shown in Fig. 7. It can be seen that by increasing the number of residual blocks from 3 to 6, the pansharpening performance is significantly improved. However, through experiments, we found that the pansharpening performance will decrease if the number of residual blocks continues to increase. The reason is that the input PAN image and the spatial component of HS image can already be finely decomposed by six residual blocks, keeping increasing the residual blocks makes the training inefficient. We also compare the fusion performance of the models with Resblocks and that of the models without Resblocks and only each level of PAN branch with six Resblocks. These results in Fig. 7 indicate that, in the dual-branch detail extraction network of the proposed DBDENet method, the PAN detail extraction branch and the detail fusion branch with six residual blocks achieve better fusion performance than other models.

2) Loss Function: We test the average quantitative assessment for different loss functions on the Pavia Center dataset.

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TABLE V
AVERAGE QUANTITATIVE ASSESSMENT RESULTS OF THE PROPOSED METHOD WITH DIFFERENT STRUCTURES ON THE PAVIA CENTER DATASET

l_1 loss	l_2 loss	Average-pooling	Max-pooling	SCAFM	SAFM	w/o SAFM	w/o FM	Dual-branch	Single-branch	Multi-level	Single-level	CC	SAM	RMSE	ERGAS
✓		✓		✓				✓		✓		**0.9704**	**5.1321**	**0.0193**	**3.0070**
	✓	✓		✓				✓		✓		0.9649	5.7093	0.0214	3.326
✓			✓	✓				✓		✓		0.9659	5.4524	0.0209	3.2585
	✓		✓	✓				✓		✓		0.9643	5.7571	0.0216	3.3636
✓		✓			✓			✓		✓		0.9660	5.4322	0.0208	3.2483
✓		✓				✓		✓		✓		0.9646	5.5335	0.0218	3.2893
✓		✓					✓		✓	✓		0.9621	6.1824	0.023	3.5327
✓		✓		✓				✓			✓	0.9663	5.6573	0.0212	3.2789

It can be seen from Table V, compared with l_2 loss function, better pansharpening performance can be achieved by l_1. This is because l1 is usually used to encourage distortion sparsity so that more structured information and details can be better preserved in the pansharpened results. Therefore, l_1 is used as the loss function in this article.

3) Average-Pooling: In order to analyze the effects of different pooling operations, we test the fusion performance of the proposed method with max-pooling and average-pooling on the Pavia Center dataset. The results of different pooling operations are shown in Table V. It can be seen that average-pooling performs better than max-pooling.

4) Spatial Cross-Attention Fusion Module: Although residual blocks are introduced in each level of the dual-branch detail extraction network to extract spatial features, it is still essential to add an SCAFM between the two branches. We use SCAFM because it can transfer important spatial features from one branch to another, thus effectively fusing the spatial information of the two branches. We divide our model structures into three types, i.e., model with SCAFM between two branches, model with spatial attention module on each branch (abbreviated as "SAFM"), and model without spatial attention module (abbreviated as "w/o SAFM"), and tune the parameters of these models to make them get the best performance. Pansharpening performance of these three models are shown in Table V. By comparing the results presented in rows 1, 5, and 6 of Table V, we observe that model "SCAFM" gets the best performance, followed by model "SAFM," and finally model "w/o SAFM." This indicates that the attention mechanism can effectively improve the fusion performance of the model, and using SCAFM between two branches can obtain better performance than applying attention masks to their own branches.

5) Dual-Branch Fusion Network: The proposed dual-branch fusion module contains two branches: the PAN detail extraction branch and the detail fusion branch. In order to prove the superiority of the dual-branch fusion module, we compared the performance of the model with the dual-branch fusion module and the model with the single-branch fusion module under the same setting of other parameters. The single-branch module contains only a PAN detail extraction branch for extracting the missing details of the HS image and does not contain a spatial attention fusion module (abbreviated as "w/o FM"). The results shown in rows 1 and 7 of Table V reveal that, compared with the dual-branch module, the single-branch module leads to a significant decrease in the pansharpening performance of the model.

6) Multilevel Structure: In order to verify the superiority of the multilevel structure, we train and test on two models with and without multilevel structure, and show the average quantitative assessment results on the Pavia Center dataset in Table V. The single-level model is constructed by removing the second level structure from the multi-level model. Both the multilevel structure and the single-level structure are all equipped with 12 Resblocks and use the same loss function and pooling operation. As can be seen, the multilevel structure achieves a better performance.

IV. CONCLUSION

In this study, we present a DBDENet framework for the task of HS and PAN images fusion, which designs a single pre-trained model for the datasets with an arbitrary number of spectral bands by fine-tuning the parameters of the last convolutional layer of the second SCAFM module. The DBDENet method designs a dual–branch fusion-based multilevel detail extraction subnetwork to achieve accurate estimation of missing details of HS image. In particular, the dual-branch fusion subnetwork contains two branches which are the PAN detail extraction branch and the detail fusion branch, and extracts the details from HS and PAN images level by level. The extracted details from the PAN and HS images are fused by an SCAFM module at each level to fully estimate the missing details. Experimental results demonstrate the validity of the proposed DBDENet method in preserving spectral information and enhancing spatial details of the original HS image. The proposed DBDENet method in this article can be applied to the pansharpening tasks where the resolution of the PAN images is 2^i times that of HS images. In the future, we will study how to extend the network to HS and PAN images with any level of scaling factors to make it more generalized.

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